Substituting dietary saturated for monounsaturated fat impairs insulin sensitivity in healthy men and women: The KANWU Study.

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AIMS/HYPOTHESIS: The amount and quality of fat in the diet could be of importance for development of insulin resistance and related metabolic disorders. Our aim was to determine whether a change in dietary fat quality alone could alter insulin action in humans. METHODS: The KANWU study included 162 healthy subjects chosen at random to receive a controlled, isoenergetic diet for 3 months containing either a high proportion of saturated (SAFA diet) or monounsaturated (MUFA diet) fatty acids. Within each group there was a second assignment at random to supplements with fish oil (3.6 g n-3 fatty acids/d) or placebo. RESULTS: Insulin sensitivity was significantly impaired on the saturated fatty acid diet (-10%, $p = 0.03$) but did not change on the monounsaturated fatty acid diet (+2%, NS) ($p = 0.05$ for difference between diets). Insulin secretion was not affected. The addition of n-3 fatty acids influenced neither insulin sensitivity nor insulin secretion. The favourable effects of substituting a monounsaturated fatty acid diet for a saturated fatty acid diet on insulin sensitivity were only seen at a total fat intake below median (37E%). Here, insulin sensitivity was 12.5% lower and 8.8% higher on the saturated fatty acid diet and monounsaturated fatty acid diet respectively ($p = 0.03$). Low density lipoprotein cholesterol (LDL) increased on the saturated fatty acid diet (+4.1%, $p < 0.01$) but decreased on the monounsaturated fatty acid diet (MUFA) (-5.2, $p < 0.001$), whereas lipoprotein (a) [Lp(a)] increased on a monounsaturated fatty acid diet by 12% ($p < 0.001$). CONCLUSIONS/INTERPRETATION: A change of the proportions of dietary fatty acids, decreasing saturated fatty acid and increasing monounsaturated fatty acid, improves insulin sensitivity but has no effect on insulin secretion. A beneficial impact of the fat quality on insulin sensitivity is not seen in individuals with a high fat intake (> 37E%).